

Laxman Sharma

📍 Kathmandu-26 Samakhushi, Nepal ✉ laxman.sharma1811@gmail.com ☎ +977-9848590208
🌐 laxman-sharma.com.np in laxmansharma1 📱 laxmansharma1811

Professional Summary

AI/ML Engineer and Full-Stack Developer with hands-on experience building and deploying production-grade LLM applications, RAG pipelines, and multi-agent systems. Designed and shipped BHASABODH (Romanized-to-Devanagari NLP chatbot) at eSewa and a live RAG PDF chatbot using LangChain, ChromaDB, and Google Gemini. Built and deployed EDUHUB Universal Services Portal serving real users in production. Passionate about applying AI, NLP, and scalable software engineering to solve real-world problems.

Education

Sunway College Kathmandu

Graduated March 2025

Bachelor of Computer Science (Honours) — CGPA: 3.59 / 4.0

Work Experience

AI/ML Engineering Intern

Kathmandu, Nepal

eSewa Ltd — Nepal's Leading Digital Wallet Platform

Nov 2024 – Mar 2025

- Developed **BHASABODH**, a production NLP system translating Romanized Nepali to Devanagari using custom fine-tuned transformer models (BLOOM, BERT variants), enabling accurate transliteration at scale
- Designed and executed end-to-end ML pipelines: data cleaning, tokenization, model fine-tuning, BLEU score evaluation, and inference optimization for reduced latency
- Deployed an interactive translation chatbot via **FastAPI** with a Tailwind CSS frontend, enabling real-time user testing and R&D feedback loops
- Collaborated cross-functionally with the R&D team to iterate on model performance, improving translation accuracy through systematic benchmarking

Data Engineering Intern

Kathmandu, Nepal

Dhytica Inc. Pvt. Ltd

Jul 2024 – Nov 2024

- Designed and implemented **ETL pipelines** and automated DAGs using **Apache Airflow** for scheduled data ingestion and transformation workflows
- Gained hands-on experience with cloud-based data infrastructure, database management, and real-time data processing using **MinIO** for scalable object storage
- Built working knowledge of data warehousing, big data technologies, and pipeline orchestration in a production data engineering environment

Projects

Multi-Agent Research System [AI AGENTS]

[Live Demo](#)

- Architected an autonomous **multi-agent pipeline** for end-to-end web research and structured report generation, demonstrating production-level agentic AI design
- Built a **Search Agent** leveraging the **Tavily API** for real-time web retrieval and a **Scraping Agent** using **BeautifulSoup** for structured content extraction from target web pages
- Integrated **Google Gemini** via **LangChain** as the core LLM for synthesizing retrieved data into coherent, high-quality research reports
- Implemented a **Critic/Evaluation Agent** to iteratively review and refine generated outputs, improving factual consistency and report quality
- Designed a modular, extensible agent architecture enabling independent agent swapping and pipeline scaling
- Technologies: Python, LangChain, Google Gemini API, Tavily API, BeautifulSoup, Multi-Agent Systems

RAG PDF Chatbot [\[DEPLOYED\]](#)

[Live Demo](#) 

- Built and deployed a **Retrieval-Augmented Generation (RAG)** system enabling natural language Q&A over user-uploaded PDF documents, with a full REST API and web frontend
- Engineered a document ingestion pipeline using **LangChain** (**PyPDFLoader**, **RecursiveCharacterTextSplitter**), **HuggingFace** sentence embeddings (**all-MiniLM-L6-v2**), and **ChromaDB** vector store for semantic retrieval
- Connected **Google Gemini 2.5 Flash** as the generative LLM via LangChain's **create_retrieval_chain**, delivering context-grounded responses with low hallucination
- Deployed on **Render** with lazy model initialization to prevent cold-start timeouts; resolved ChromaDB/SQLite compatibility via **pysqlite3-binary** hot-swap
- Technologies: Python, FastAPI, LangChain, ChromaDB, HuggingFace Transformers, Google Gemini API, Render

EDUHUB Universal Services Portal [\[LIVE\]](#)

[Live Site](#) 

- Designed, developed, and deployed a production **Django** education portal from the ground up, serving real users seeking university and scholarship information
- Provisioned a **VPS** (Nest Nepal) with **Nginx** + **Uvicorn**, configured **PostgreSQL** for data persistence, and ensured secure, high-availability uptime
- Implemented mobile-first responsive UI, SEO metadata optimization, and page-speed tuning to improve organic search rankings and user engagement
- Technologies: Python, Django 4.2, PostgreSQL, HTML/CSS, Nginx, Uvicorn, VPS, SEO

Ecommerce Product Aggregator

[GitHub](#) 

- Built a full-featured e-commerce platform with product listing, user authentication, historical price trend visualization, and analytics dashboard
- Technologies: Django, Tailwind CSS, PostgreSQL

Automated Web Scraping Toolkit

[GitHub](#) 

- Developed a **Map Scraper** with automated navigation, pagination handling, and structured data extraction; built an **Amazon Scraper** for large-scale product detail collection
- Technologies: Python, Selenium, Requests, BeautifulSoup

Technical Skills

Languages: Python (Advanced), Java, JavaScript, TypeScript, SQL, HTML5, CSS3

AI/ML & LLMs: PyTorch, TensorFlow, Transformers, BERT/BLOOM, scikit-learn, pandas, NumPy, LangChain, RAG, Multi-Agent Systems, ChromaDB, HuggingFace, Google Gemini API, Tavily API

Web & API Development: Django, FastAPI, REST APIs, Jinja2

Data Engineering: Apache Airflow, ETL Pipelines, MinIO, PostgreSQL

DevOps & Tools: Docker, Git, CI/CD, Linux, Nginx, Render, Streamlit, Selenium, Postman, JIRA, Agile/Scrum